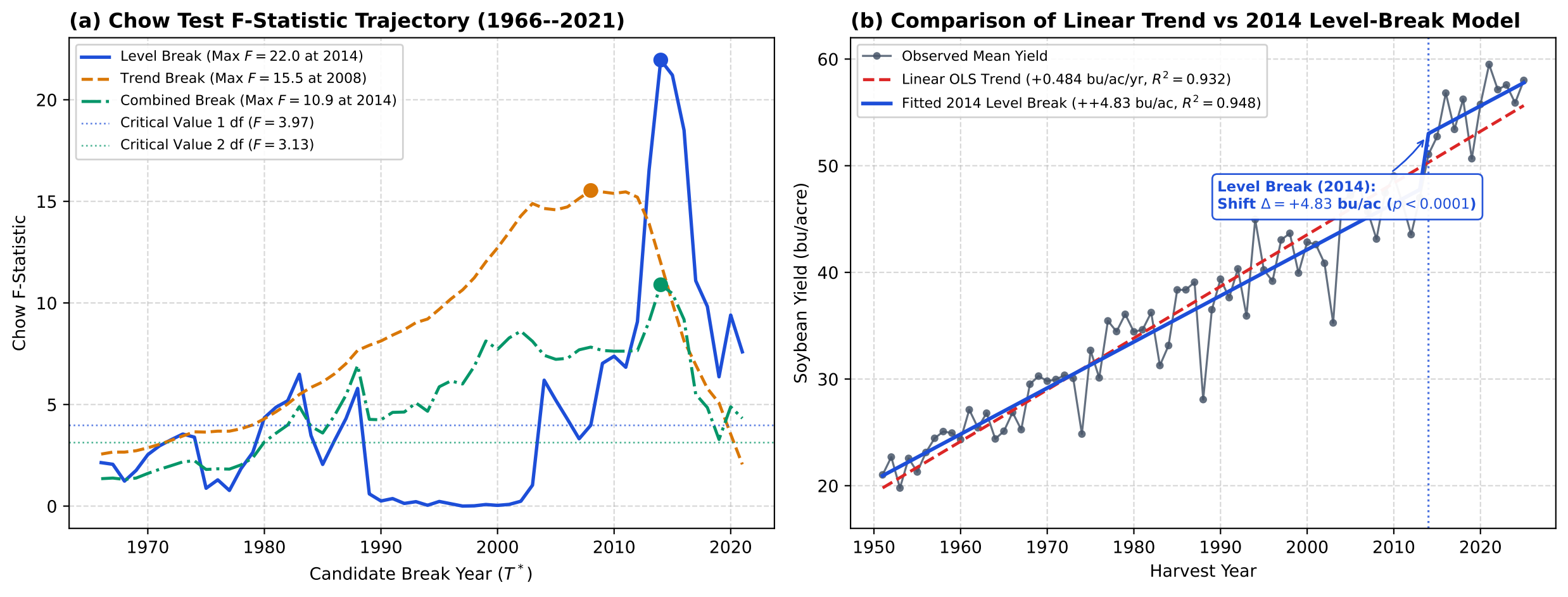
(a) Chow Test F-Statistic Trajectory (1966--2021)
Level Break (Max $F = 22.0$ at 2014)
Trend Break (Max $F = 15.5$ at 2008)
Combined Break (Max $F = 10.9$ at 2014)
Critical Value 1 df ($F = 3.97$)
Critical Value 2 df ($F = 3.13$)
Chow F-Statistic
Candidate Break Year (T^*)
(b) Comparison of Linear Trend vs 2014 Level-Break Model
Observed Mean Yield
Linear OLS Trend (+0.484 bu/ac/yr, $R^2 = 0.932$)
Fitted 2014 Level Break (++4.83 bu/ac, $R^2 = 0.948$)
Level Break (2014):
Shift $\Delta = +4.83$ bu/ac ($p < 0.0001$)
Soybean Yield (bu/acre)
Harvest Year